



Urban Informatics
Assignment - Week 4

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 16

Total marks: (12 X 1) + (4 X 2) = 20 marks

MCQ/MSQ Question

QUESTION 1: [1 mark]

Which of the following is NOT TRUE about Procedure-Oriented Programming (POP)?

- a. Data is often global and shared across functions.
- b. Global data are vulnerable to inadvertent changes by functions.
- c. External data structure changes require revision of many dependent functions.
- d. Data is fully encapsulated and cannot be accessed directly by functions.

Correct Answer: d. Data is fully encapsulated and cannot be accessed directly by functions.

Detailed explanation: Refer to lecture 19, slide 2. Option d. describes Object-Oriented Programming, not POP.

QUESTION 2: [1 mark]

Which of the following is NOT a Python library?

- a. SciPy
- b. Spyder
- c. Matplotlib
- d. NumPy

Correct Answer: b. Spyder

Detailed explanation: Refer to lecture 16. Spyder (Scientific Python Development Environment) is an Integrated Development Environment.

QUESTION 3: [1 mark]

A variable that is defined inside a function and can be accessed only within that function is called _____ variable.

- a. Global
- b. String
- c. Float
- d. Local

Correct Answer: d. Local

Detailed explanation: Refer to lecture 17, slide 3.



QUESTION 4: [2 marks]

Consider the following lines of code. Assume all necessary standard libraries are imported.

```
class PublicSpace:
    def __init__(self, name):
        self.name = name

    def show_name(self):
        print(self.name)

class Park(PublicSpace):
    def __init__(self, name, area):
        super().__init__(name)
        self.area = area

park1 = Park("City Park", 20)
park1.show_name()
```

What will be the output?

- a. City Park
- b. 20
- c. Error
- d. name

Correct Answer: a. City Park

Detailed explanation: Refer to lecture 20, Inheritance in Python.

PublicSpace is a parent class that stores the common attribute *name* and defines the method *show_name()*. *Park* is a child class that inherits from *PublicSpace* and adds its own attribute *area*. *super().__init__(name)* calls the parent class constructor to initialize *name*. The object *park1* can access the parent method *show_name()*, so it prints City Park.



QUESTION 5: [2 marks]

Match the following Python libraries to the functions they are primarily used for.

Library	Primary function
P. SciPy	i. Machine learning algorithms for classification, regression, and clustering
Q. Statsmodels	ii. End-to-end platform for machine learning and deep learning
R. Scikit-Learn	iii. Scientific computing for optimization, integration, and numerical methods
S. TensorFlow	iv. Statistical modeling, estimation, and hypothesis testing

- a. P-iii; Q-iv; R-i; S-ii
- b. P-iv; Q-ii; R-iii; S-i
- c. P-i; Q-iv; R-ii; S-iii
- d. P-ii; Q-i; R-iv; S-iii

Correct Answer: a. P-iii; Q-iv; R-i; S-ii

Detailed explanation: Refer to lecture 16, slide 8 and 9.

QUESTION 6: [1 mark]

Why is calling one function from another considered a good programming practice?

- a. It increases execution time
- b. It avoids syntax errors
- c. It reduces code duplication and improves reusability
- d. It removes the need for variables

Correct Answer: c. It reduces code duplication and improves reusability

Detailed Solution: Refer to lecture 17, slide 9.

QUESTION 7: [1 mark]

Properties of Object-Oriented Programming (OOP) are-

- A. Compilation
- B. Libraries
- C. Inheritance
- D. Polymorphism
- E. Dynamic Binding
- F. Message Passing

- a. A, B, C and D
- b. A, C, E and F
- c. B, C, D and E
- d. C, D, E and F

Correct Answer: d. C, D, E and F



Detailed Solution: Refer to lecture 19.

QUESTION 8: [1 mark]

Consider the following lines of code. Assume all necessary standard libraries are imported.

```
def add(a, b):  
    return a + b  
  
def calculate():  
    result = add(3, 4)  
    return result  
  
print(calculate())
```

What will be the output?

- a. add (3, 4)
- b. 7
- c. 3+4
- d. error

Correct Answer: b. 7

Detailed Solution: Refer to lecture 17, function calling a function.

The function *calculate()* calls the function *add(3, 4)*. The *add()* function returns the sum 7, which is stored in the variable *result*. Since *calculate()* returns *result*, the *print(calculate())* statement prints 7.

QUESTION 9: [1 mark]

Which of the following statements correctly describe features of Object-Oriented Programming (OOP)?

- A. OOP allows data to be encapsulated within objects, protecting it from accidental external modification.
 - B. OOP organizes code using objects that combine data and behavior
 - C. OOP programs cannot use inheritance or reuse existing code
 - D. OOP enables modular and maintainable program design through classes and objects
-
- a. A, B and C
 - b. A, B and D
 - c. A, C and D
 - d. B, C and D

Correct Answer: b. A, B and D

Detailed Solution: Refer to lecture 19.



QUESTION 10: [1 mark]

Consider the following lines of code. Assume all necessary standard libraries are imported.

```
.  
count = 1  
total = 0  
  
while count < 5:  
    total += count  
    count += 2  
  
print(total)
```

What will be the output?

- a. 4
- b. 6
- c. 9
- d. Infinite loop

Correct Answer: a. 4

Detailed Solution: Refer to lecture 18.

The loop runs while $count < 5$.

Iteration 1: $count = 1$; $total = 1$

Iteration 2: $count = 3$; $total = 4$

Iteration 3: $count = 5$; condition becomes False. That makes $total = 4$, not 6.

QUESTION 11: [2 marks]

Which of the following statement is CORRECT?

- a. *List* and *Tuple* are both mutable and ordered.
- b. *Array* and *List* can store elements of different data types.
- c. *Dictionary* is mutable and stores data as key–value pairs, while *Set* does not allow duplicate elements.
- d. *Tuple* and *Set* both preserve insertion order.

Correct Answer: c. *Dictionary* is mutable and stores data as key–value pairs, while *Set* does not allow duplicate elements.

Detailed Solution: Refer to lecture 18.



QUESTION 12: [1 mark]

Consider the following lines of code. Assume all necessary standard libraries are imported.

```
class Zone:
    road_width = 20

    def __init__(self, name):
        self.name = name

z1 = Zone("Market")
z2 = Zone("Residential")

z1.road_width = 30

print(z1.road_width)
print(z2.road_width)
print(Zone.road_width)
```

What will be the output?

- a. 30
20
20
- b. 30
30
30
- c. 20
20
30
- d. 20
20
20

Correct Answer: a. 30
20
20

Detailed Solution: Refer to lecture 20. `road_width` is a class variable. Assigning `z1.road_width = 30` creates an instance variable. `z2` still uses the class variable. The class itself remains unchanged.



QUESTION 13: [1 mark]

A city transport system models **Bus**, **Metro**, and **Bicycle** as separate classes. Each class has a method named *travel_time()*, but the method calculates time differently for each mode.

Which Object-Oriented Programming concept is illustrated here?

- a. Encapsulation
- b. Inheritance
- c. Polymorphism
- d. Abstraction

Correct Answer: c. Polymorphism

Detailed Solution: Refer to lecture 19. Polymorphism allows the same method name to behave differently depending on the object.

QUESTION 14: [2 marks]

Match the following Object Oriented Programming (OOP) concepts with their description.

Concept	Description
P. Encapsulation	i. Creating new classes by reusing properties of existing classes
Q. Inheritance	ii. Hiding implementation details and showing only essential features
R. Polymorphism	iii. Wrapping data and methods together and restricting direct access
S. Abstraction	iv. Using the same method name to perform different behaviors

- a. P-ii; Q-iv; R-i; S-iii
- b. P-iii; Q-i; R-iv; S-ii
- c. P-ii; Q-iii; R-i; S-iv
- d. P-i; Q-iv; R-ii; S-iii

Correct Answer: b. P-iii; Q-i; R-iv; S-ii

Detailed Solution: Refer to lecture 19.

QUESTION 15: [1 mark]

A _____ is a single Python file that organizes code and may contain functions, classes, or variables.

- a. package
- b. library
- c. module
- d. directory

Correct Answer: c. module

Detailed Solution: Refer to lecture 16.



QUESTION 16: [1 mark]

In object-oriented programming, _____ allows the method to be decided at runtime, not at compile time.

- a. method overloading
- b. dynamic binding
- c. message passing
- d. encapsulation

Correct Answer: b. dynamic binding

Detailed Solution: Refer to lecture 19.

*****END*****